

Evaluating Causal Reasoning in MLLMs for Brain Imaging

Dry Run · 5 Models · 3 Neuroimaging Cases · 30 Questions

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The Problem with Accuracy

Standard benchmarks ask: did the model get the right answer?

- Did the model anchor its reasoning to what it saw in the image?
- Did its justification actually lead to the answer it gave?
- Did it build on what it found in earlier reasoning phases?

A model can score 100% accuracy and still be clinically unsafe.

This framework evaluates the reasoning chain, not just the endpoint.

The Framework: Causal Reasoning Chain

Phase	Abbrev.	Failure Type	Clinical Task
Anatomical & Imaging Assessment	AIA	Perception	Identify modality, anatomy, structures
Lesion Identification & Localization	LIL	Perception	Detect and localize pathology
Diagnostic Synthesis & Causal Reasoning	DSCR	Reasoning	Reach a diagnosis, explain causally
Prognostic Judgment & Risk Forecasting	PJRF	Decision	Grade severity, predict outcome
Therapeutic Cycle Management	TCM	Decision	Recommend treatment and follow-up

Gating rule: Model must score $\geq 50\%$ on phase N to unlock phase N+1.

Gate failure → downstream phases are blocked, not attempted.

Three Faithfulness Dimensions

① Local Faithfulness

Does the model's stated reasoning actually support the answer it gave?

Second API call: given only the reasoning, predict the answer. Divergence = unfaithful.

→ *Catches answers chosen independently of the reasoning*

② Grounding Faithfulness

Does the reasoning anchor to imaging features present in the case?

Fraction of expected visual terms cited (e.g. ring enhancement, midline shift).

→ *Catches reasoning disconnected from the image*

③ Chain Faithfulness

Does each phase build on conclusions from prior phases?

Does DSCR reference AIA/LIL findings? Does TCM reference DSCR and PJRF?

→ *Catches reasoning that doesn't accumulate*

These three dimensions are independent of accuracy. A model can be correct on all — or correct on none.

Core Finding: Grounding Faithfulness Collapses at Decision Phases

Every model shows the same pattern — strong grounding at perception, systematic collapse by TCM.

Model	AIA	LIL	DSCR	PJRF	TCM
llama3.2-3b	73%	63%	54%	31%	22%
gpt-oss-20b	73%	73%	64%	24%	16%
llama3.1-8b	73%	—	—	—	10%

Grounding Faithfulness (higher = model is anchoring reasoning to imaging features)

Models stop anchoring to imaging features precisely at the phases where clinical safety depends on it most — diagnosis, prognosis, treatment.

This is the hypothesis from the proposal. The dry run validates it holds across all tested models.

Accuracy Hides the Problem

Model	Accuracy	Chain Complete	Grounding @ TCM	Correct-but-Unfaithful
llama3.1-8b	100%	Yes	10%	1 instance
gpt-oss-20b	93.3%	Yes	16%	0
llama3.2-3b	93.3%	Yes	22%	1 instance
qwen2.5-7b	93.3%	Yes	—	0
deepseek-r1-7b	77.8%	33%	—	0

Standard benchmarks would pass llama3.1-8b and gpt-oss-20b. This framework flags both.

**gpt-oss-20b is the strongest local model (20B, perfect local faithfulness, zero hallucinated correctness), but still loses clinical grounding at TCM (16%).
A capable model failing this way is the finding.**

Gating: Exposing Hidden Reasoning Failures

deepseek-r1-7b — a chain-of-thought specialist

- Scored 100% at AIA and LIL (perception phases — no problem here)
- Blocked at DSCR on 2 out of 3 cases → 0% accuracy at the reasoning phase
- Chain completion rate: 33%

Without gating, this model would appear competent.
The framework surfaced a hard reasoning failure that flat accuracy cannot see.

The gating mechanism works as designed.

Dry Run Scope and Limitations

What was tested

- 5 models (3B – 20B parameters)
- 3 synthetic neuroimaging cases
- 30 questions total, text-based

What this doesn't cover yet

- Actual images — text only
- Statistical generalizability (3 cases)
- Frontier models (GPT-4o, Claude, Gemini)
- Human clinician baseline

3 cases is sufficient to validate the framework mechanics — not to make statistical claims. The instrument works. Scale is next.

Moving Forward

1

Dataset Access

OmniBrainBench or real neuroimaging cases to scale from 3 cases to a full benchmark set.

2

Clinician Baseline Direction

Who to involve for annotation, what the protocol looks like, whether IRB is required.